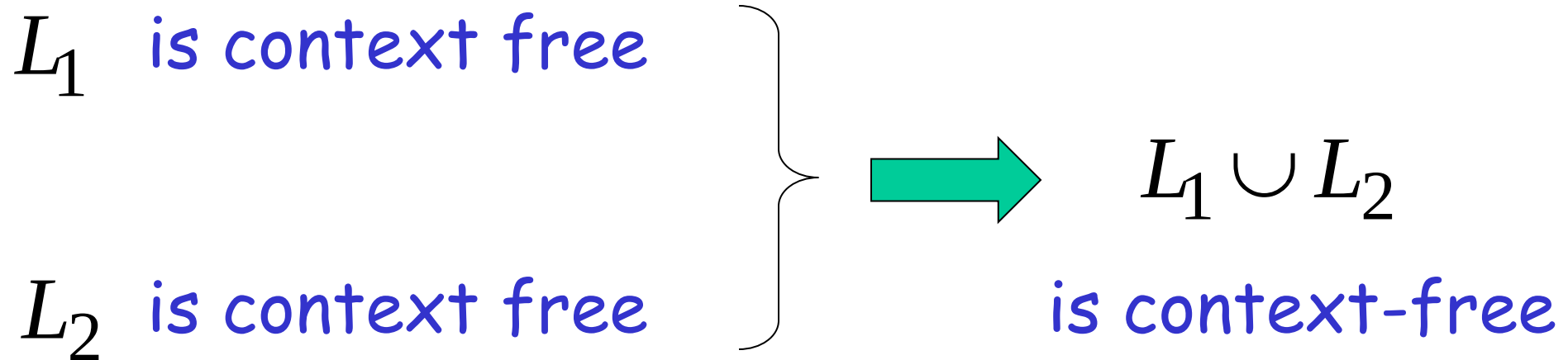


# Properties of Context-Free languages

# Union

Context-free languages  
are closed under: **Union**



# Example

Language

Grammar

$$L_1 = \{a^n b^n\}$$

$$S_1 \rightarrow aS_1b \mid \lambda$$

$$L_2 = \{ww^R\}$$

$$S_2 \rightarrow aS_2a \mid bS_2b \mid \lambda$$

Union

$$L = \{a^n b^n\} \cup \{ww^R\}$$

$$S \rightarrow S_1 \mid S_2$$

In general:

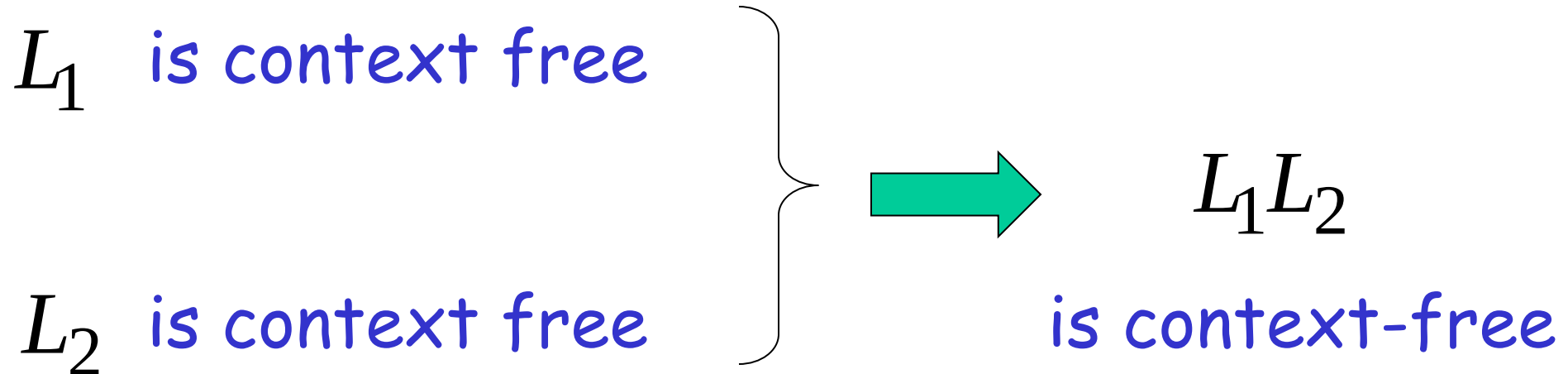
|                            |            |
|----------------------------|------------|
| For context-free languages | $L_1, L_2$ |
| with context-free grammars | $G_1, G_2$ |
| and start variables        | $S_1, S_2$ |

|                                 |                              |
|---------------------------------|------------------------------|
| The grammar of the <b>union</b> | $L_1 \cup L_2$               |
| has new start variable          | $S$                          |
| and additional production       | $S \rightarrow S_1 \mid S_2$ |

# Concatenation

Context-free languages  
are closed under:

**Concatenation**



# Example

Language

Grammar

$$L_1 = \{a^n b^n\}$$

$$S_1 \rightarrow aS_1b \mid \lambda$$

$$L_2 = \{ww^R\}$$

$$S_2 \rightarrow aS_2a \mid bS_2b \mid \lambda$$

## Concatenation

$$L = \{a^n b^n\} \{ww^R\}$$

$$S \rightarrow S_1 S_2$$

In general:

|                            |            |
|----------------------------|------------|
| For context-free languages | $L_1, L_2$ |
| with context-free grammars | $G_1, G_2$ |
| and start variables        | $S_1, S_2$ |

|                                         |                         |
|-----------------------------------------|-------------------------|
| The grammar of the <b>concatenation</b> | $L_1 L_2$               |
| has new start variable                  | $S$                     |
| and additional production               | $S \rightarrow S_1 S_2$ |

# Star Operation

Context-free languages  
are closed under:

**Star-operation**

$L$  is context free   $L^*$  is context-free



# Example

Language

Grammar

$$L = \{a^n b^n\}$$

$$S \rightarrow aSb \mid \lambda$$

## Star Operation

$$L = \{a^n b^n\}^*$$

$$S_1 \rightarrow SS_1 \mid \lambda$$

In general:

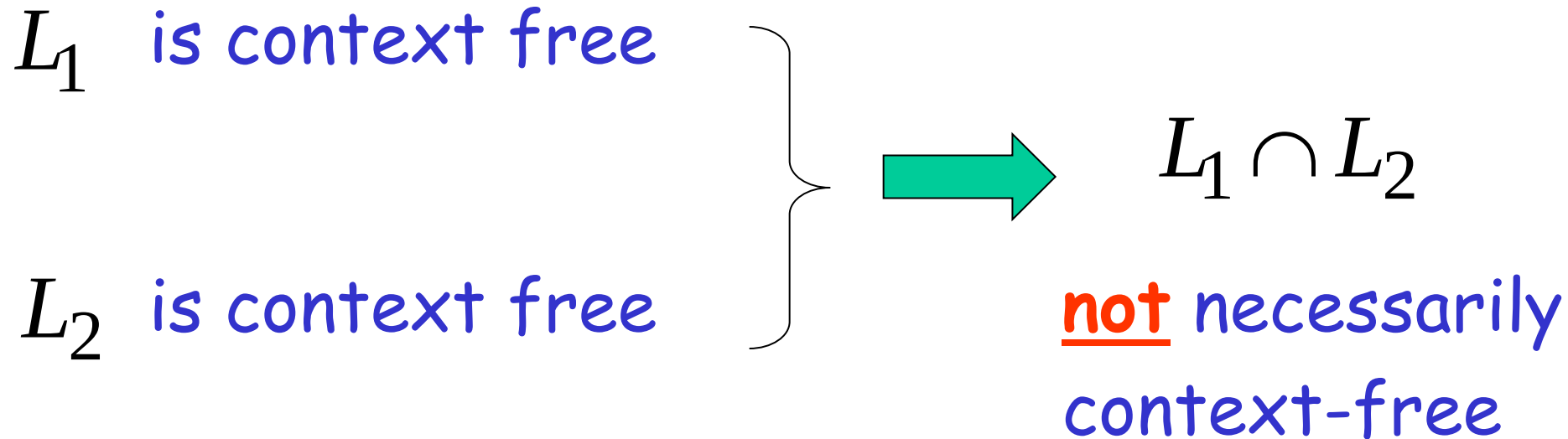
|                           |     |
|---------------------------|-----|
| For context-free language | $L$ |
| with context-free grammar | $G$ |
| and start variable        | $S$ |

|                                          |                                     |
|------------------------------------------|-------------------------------------|
| The grammar of the <b>star operation</b> | $L^*$                               |
| has new start variable                   | $S_1$                               |
| and additional production                | $S_1 \rightarrow SS_1 \mid \lambda$ |

# Negative Properties of Context-Free Languages

# Intersection

Context-free languages  
are not closed under: **intersection**



# Example

$$L_1 = \{a^n b^n c^m\}$$

Context-free:

$$S \rightarrow AC$$

$$A \rightarrow aAb \mid \lambda$$

$$C \rightarrow cC \mid \lambda$$

$$L_2 = \{a^n b^m c^m\}$$

Context-free:

$$S \rightarrow AB$$

$$A \rightarrow aA \mid \lambda$$

$$B \rightarrow bBc \mid \lambda$$

Intersection

$$L_1 \cap L_2 = \{a^n b^n c^n\} \quad \text{NOT context-free}$$

# Complement

Context-free languages  
are not closed under:

complement

$L$  is context free  $\longrightarrow \overline{L}$  not necessarily  
context-free

# Example

$$L_1 = \{a^n b^n c^m\}$$

$$L_2 = \{a^n b^m c^m\}$$

Context-free:

$$S \rightarrow AC$$

$$A \rightarrow aAb \mid \lambda$$

$$C \rightarrow cC \mid \lambda$$

Context-free:

$$S \rightarrow AB$$

$$A \rightarrow aA \mid \lambda$$

$$B \rightarrow bBc \mid \lambda$$

Complement

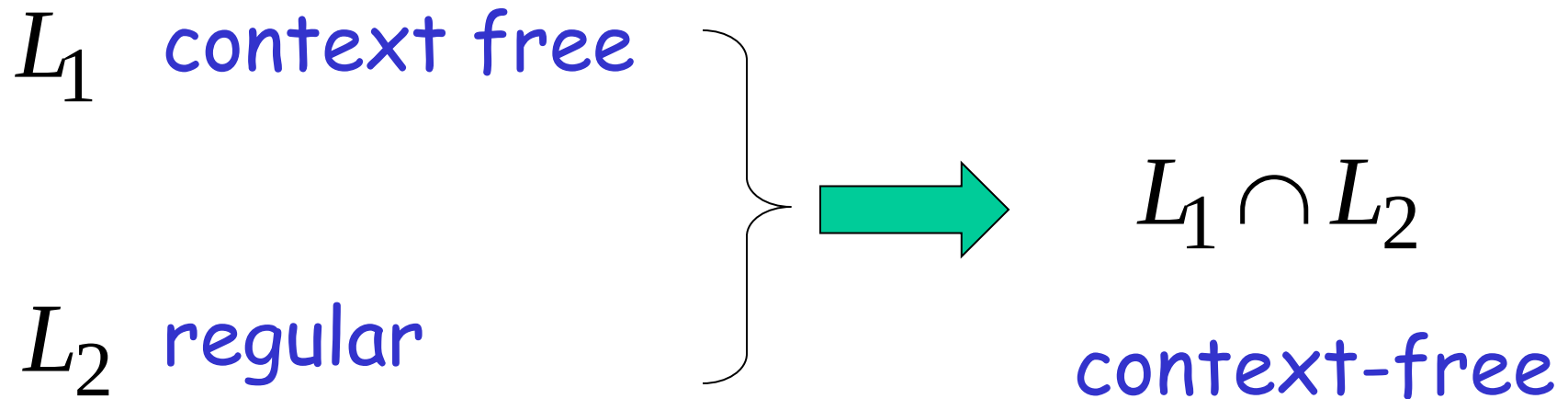
$$\overline{\overline{L_1} \cup \overline{L_2}} = L_1 \cap L_2 = \{a^n b^n c^n\}$$

NOT context-free

# Intersection of Context-free languages and Regular Languages



The intersection of  
a context-free language and  
a regular language  
is a context-free language



Machine  $M_1$

NPDA for  $L_1$   
context-free

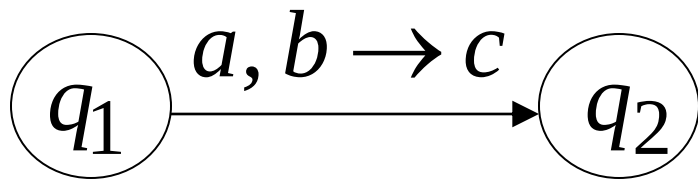
Machine  $M_2$

DFA for  $L_2$   
regular

Construct a new NPDA machine  $M$   
that accepts  $L_1 \cap L_2$

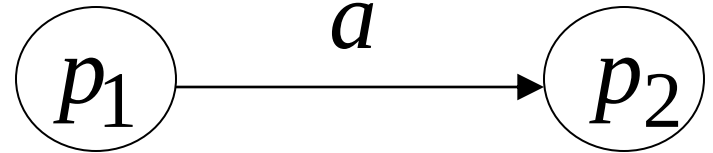
$M$  simulates in parallel  $M_1$  and  $M_2$

NPDA  $M_1$

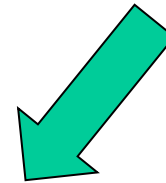


transition

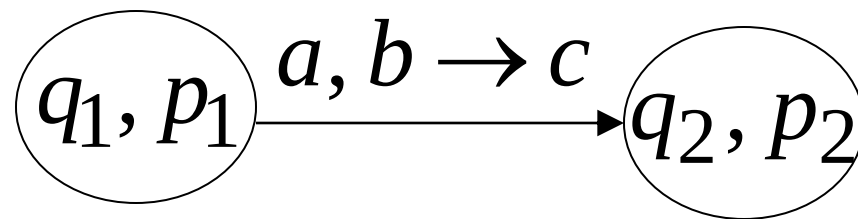
DFA  $M_2$



transition

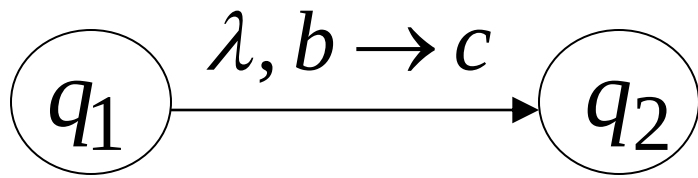


NPDA  $M$



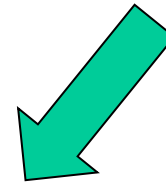
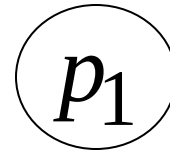
transition

NPDA  $M_1$

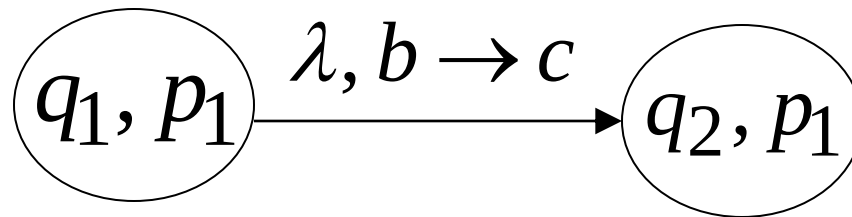


transition

DFA  $M_2$

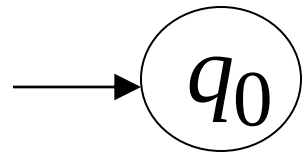


NPDA  $M$



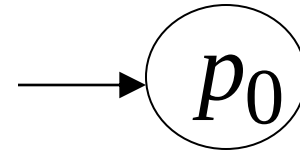
transition

NPDA  $M_1$

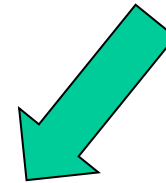


initial state

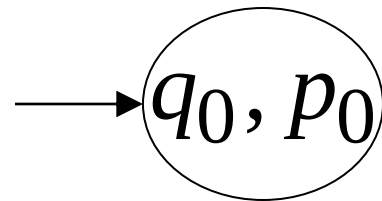
DFA  $M_2$



initial state

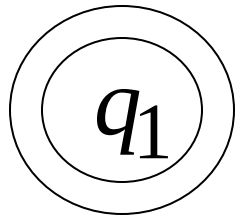


NPDA  $M$



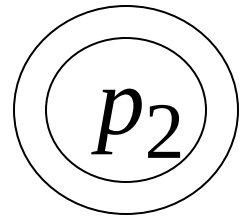
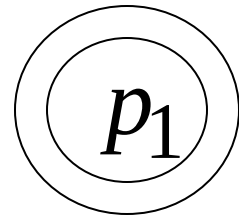
Initial state

NPDA  $M_1$



final state

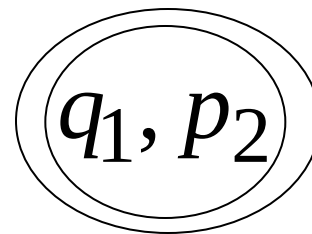
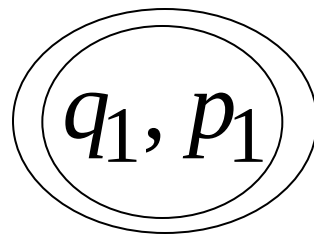
DFA  $M_2$



final states



NPDA  $M$



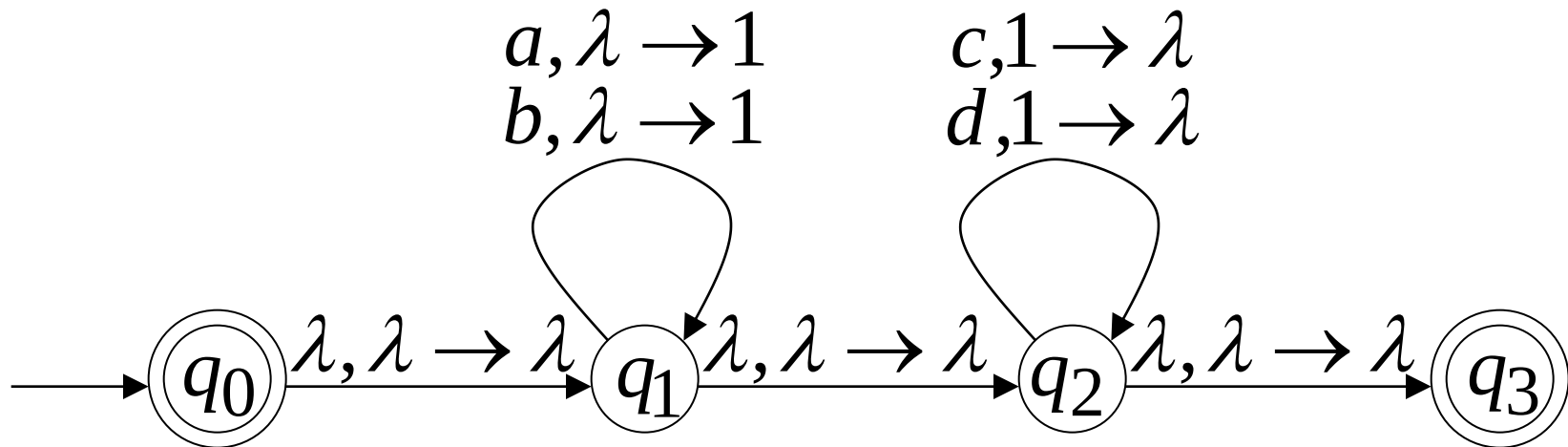
final states

**Example:**

context-free

$$L_1 = \{w_1 w_2 : |w_1| = |w_2|, w_1 \in \{a, b\}^*, w_2 \in \{c, d\}^*\}$$

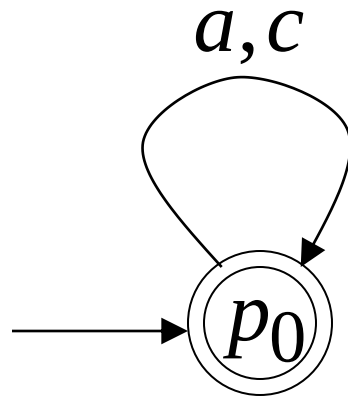
NPDA  $M_1$



regular

$$L_2 = \{a, c\}^*$$

DFA  $M_2$

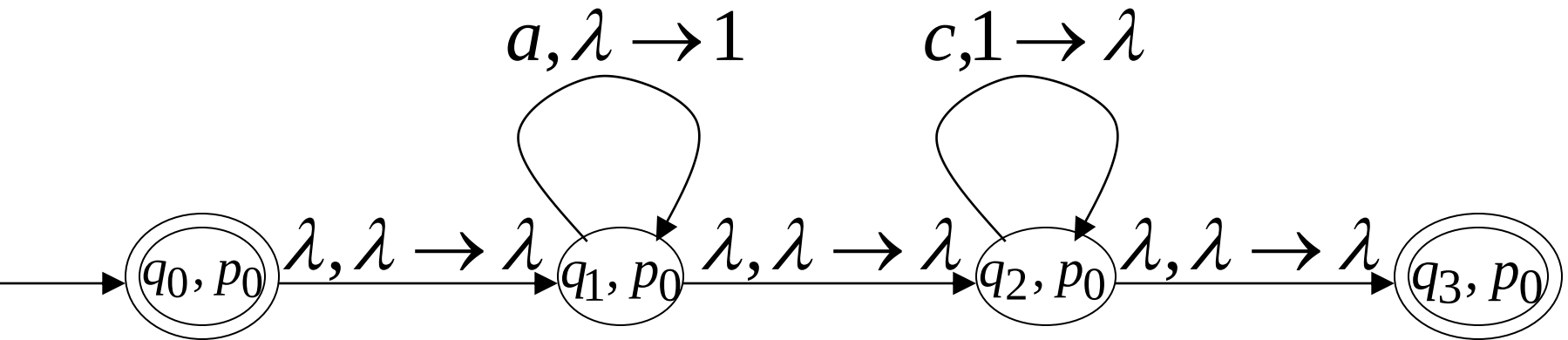




context-free

Automaton for:  $L_1 \cap L_2 = \{a^n c^n : n \geq 0\}$

NPDA  $M$



## In General:

$M$  simulates in parallel  $M_1$  and  $M_2$

$M$  accepts string  $w$  if and only if

$M_1$  accepts string  $w$  and

$M_2$  accepts string  $w$

$$L(M) = L(M_1) \cap L(M_2)$$

Therefore:

$M$  is NPDA



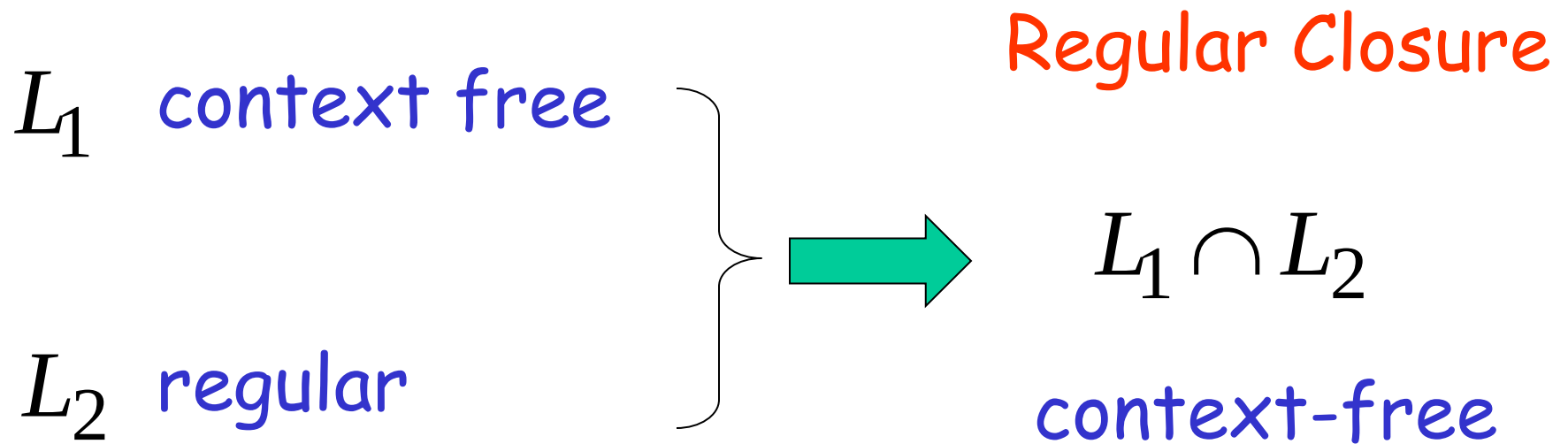
$L(M_1) \cap L(M_2)$  is context-free



$L_1 \cap L_2$  is context-free

# Applications of Regular Closure

The intersection of  
a context-free language and  
a regular language  
is a context-free language



# An Application of Regular Closure

Prove that:  $L = \{a^n b^n : n \neq 100, n \geq 0\}$

is context-free

We know:

$\{a^n b^n : n \geq 0\}$  is context-free

We also know:

$L_1 = \{a^{100}b^{100}\}$  is regular



$\overline{L_1} = \{(a + b)^*\} - \{a^{100}b^{100}\}$  is regular

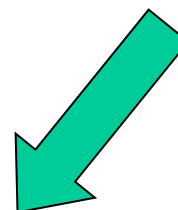
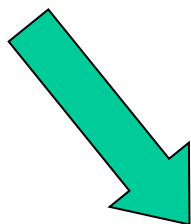


$$\{a^n b^n\}$$

context-free

$$\overline{L_1} = \{(a+b)^*\} - \{a^{100}b^{100}\}$$

regular



(regular closure)

$$\{a^n b^n\} \cap \overline{L_1}$$

context-free



$$\{a^n b^n\} \cap \overline{L_1} = \{a^n b^n : n \neq 100, n \geq 0\} = L$$

is context-free

# Another Application of Regular Closure

Prove that:  $L = \{w : n_a = n_b = n_c\}$   
is **not** context-free

If  $L = \{w : n_a = n_b = n_c\}$  is context-free

(regular closure)

Then  $L \cap \{a^*b^*c^*\} = \{a^n b^n c^n\}$

context-free

regular

context-free

**Impossible!!!**

Therefore,  $L$  is **not** context free